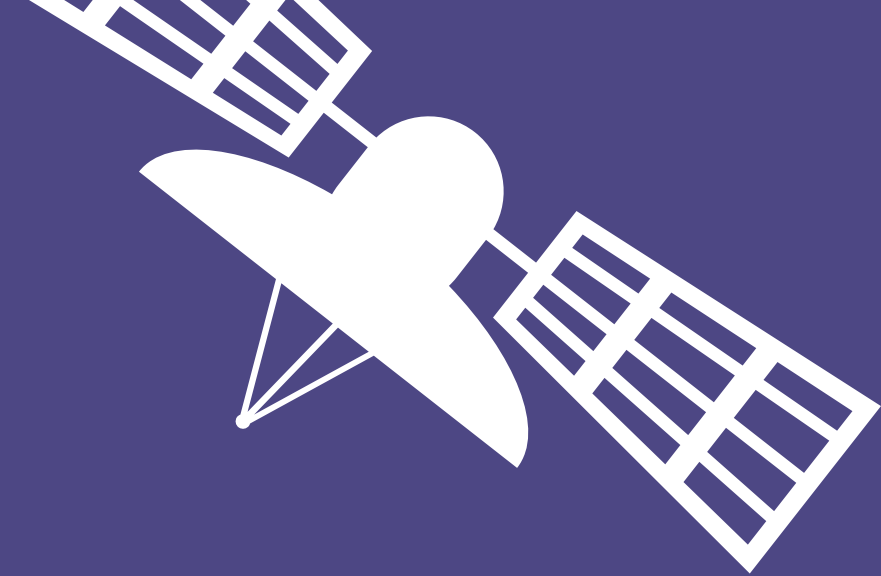




COMPANY NETWORK DESIGN



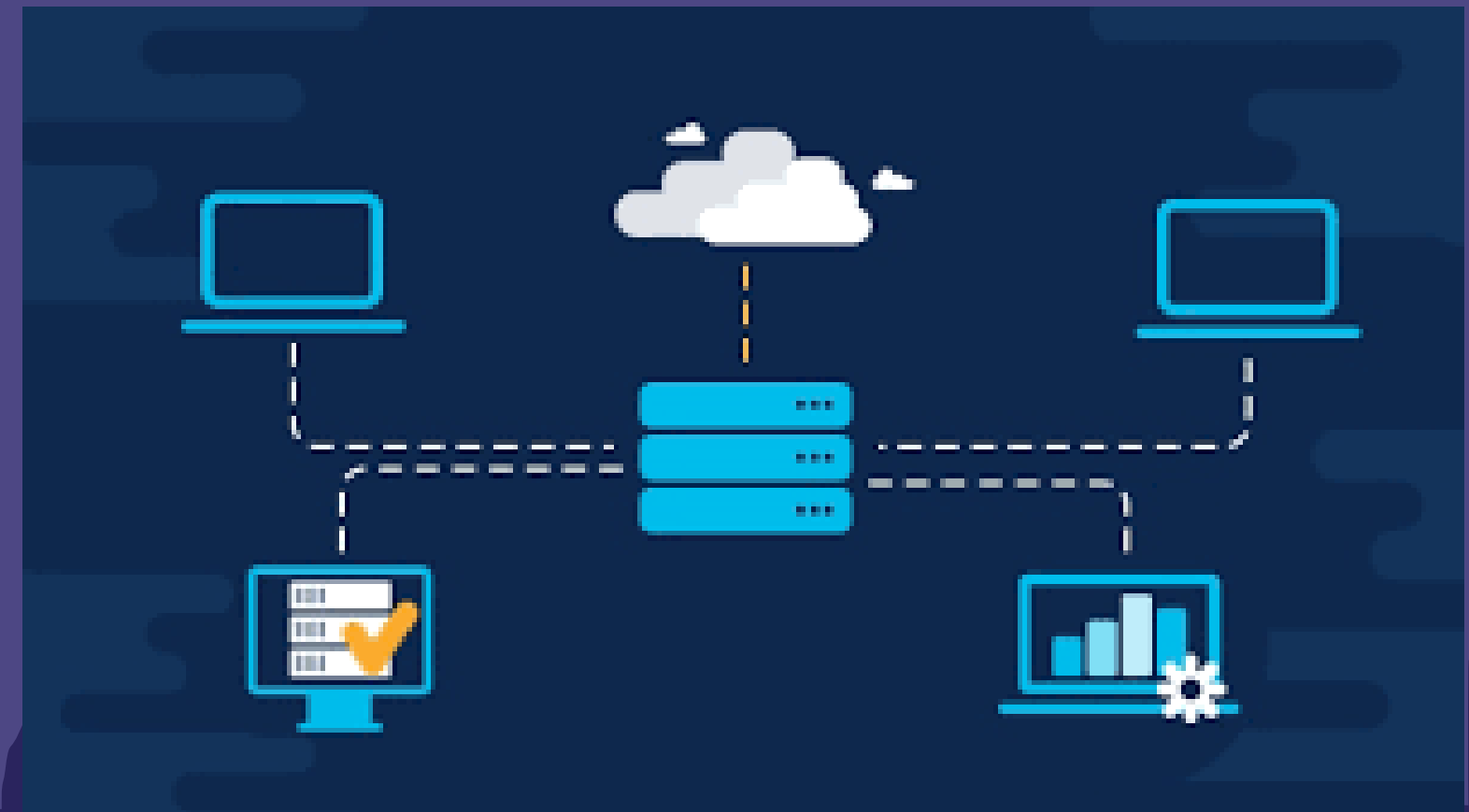
MEMBERS

- 1 Abdelrahman Wahid Abdallah
- 2 Abdelrahman Mamdouh Mohamed
- 3 Aya Mohamed Sayed
- 4 Mohamed Wahid Abdallah
- 5 Yassin Ahmad Nasr

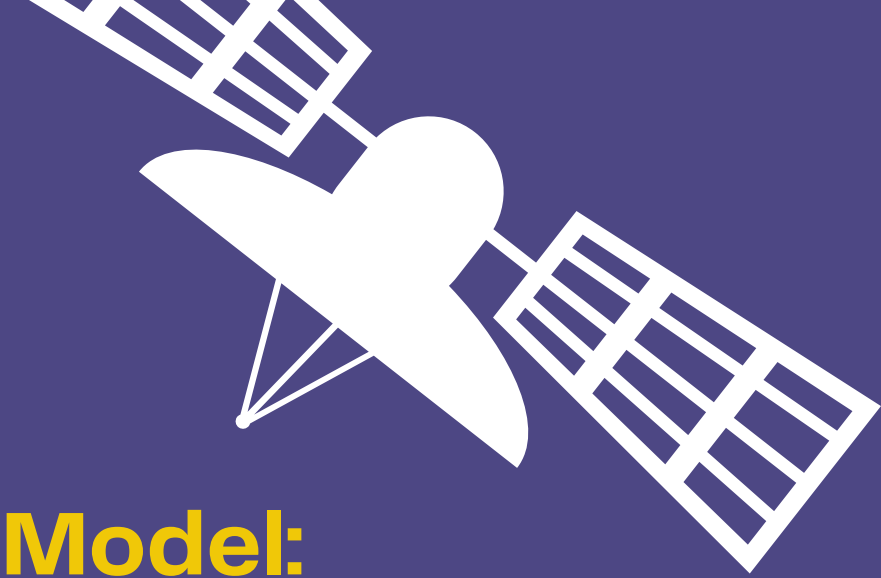


PROJECT OBJECTIVES

- Designing a secure and efficient network for a company.
 - Ensuring seamless communication between different departments.
 - Implementing backup mechanisms for uninterrupted service.
- using cisco packet tracer.**



KEY CONFIGURATIONS



- **IP Addressing:**

Subnetting for efficient use of IP addresses.
Use public IPs for internet access. (PAT)

- **DHCP Configuration:**

Automatic IP assignment to devices.

- **DNS Server Configuration:**

Automatic DNS assignment to devices.

- **Hierarchical Model:**

Two routers for redundancy.

Multilayer switches for
department connections.

- **VLANs (Virtual LANs):**

Separate networks for each
department.

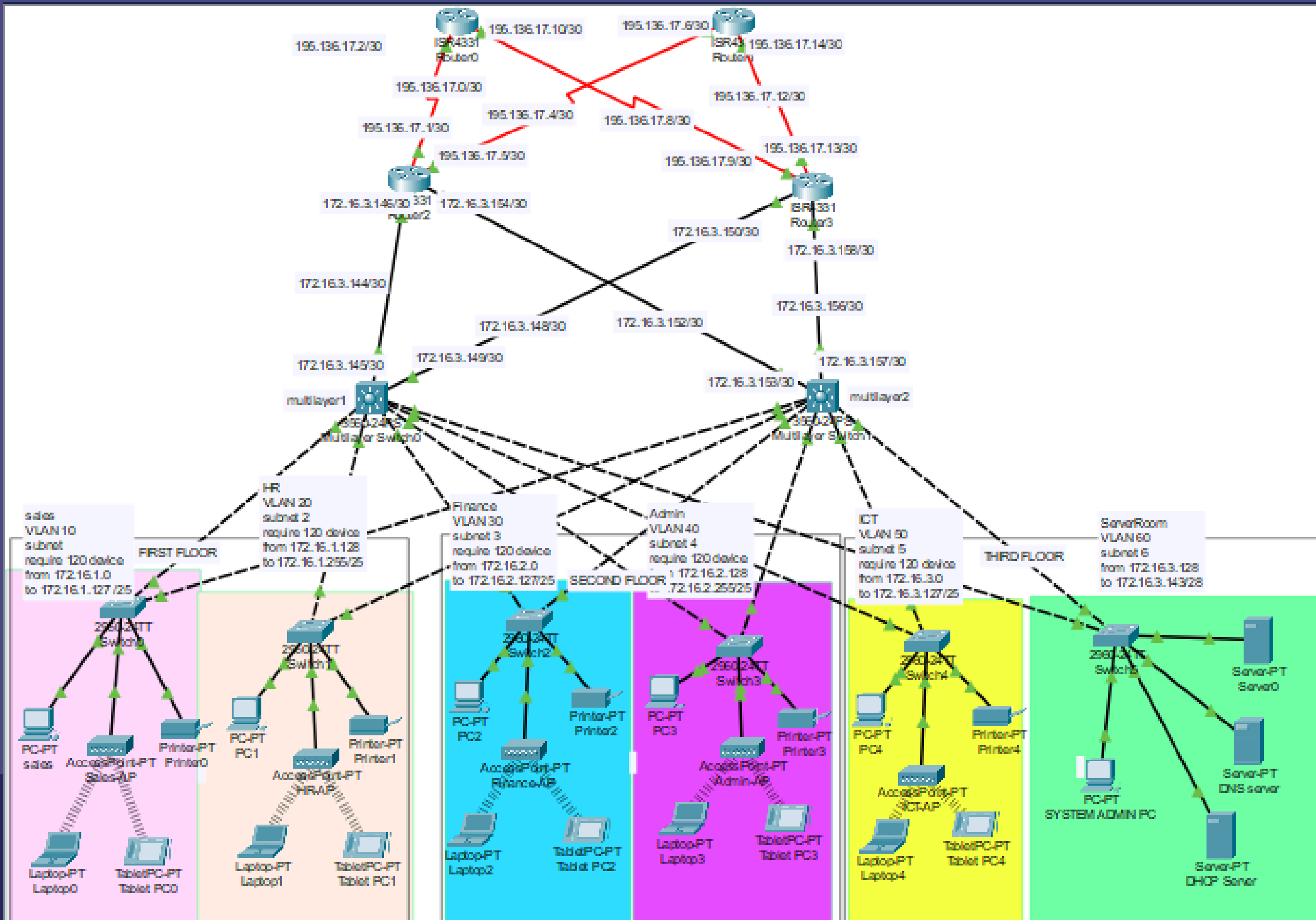


ADDITIONAL SECURITY FEATURES

- **SSH (Secure Shell):**
 - Used to secure remote access to routers and switches
 - Protects configuration data through encryption.
- **Port Security:**
 - Applied on switch ports to block unauthorized devices from connecting
 - Ensures only trusted devices access the network.



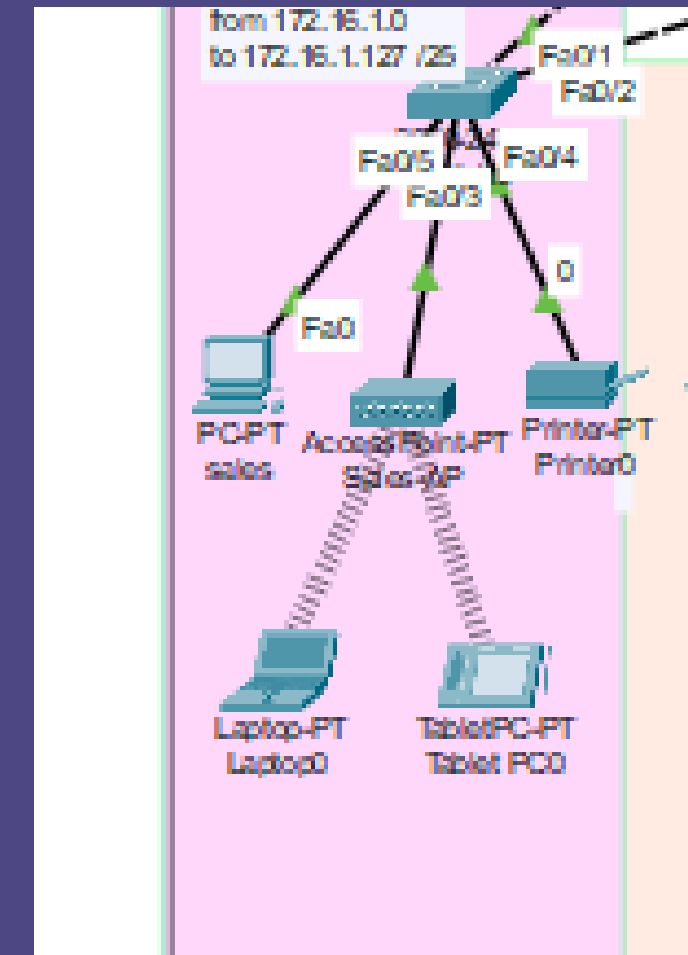
COMPANY NETWORK DESIGN



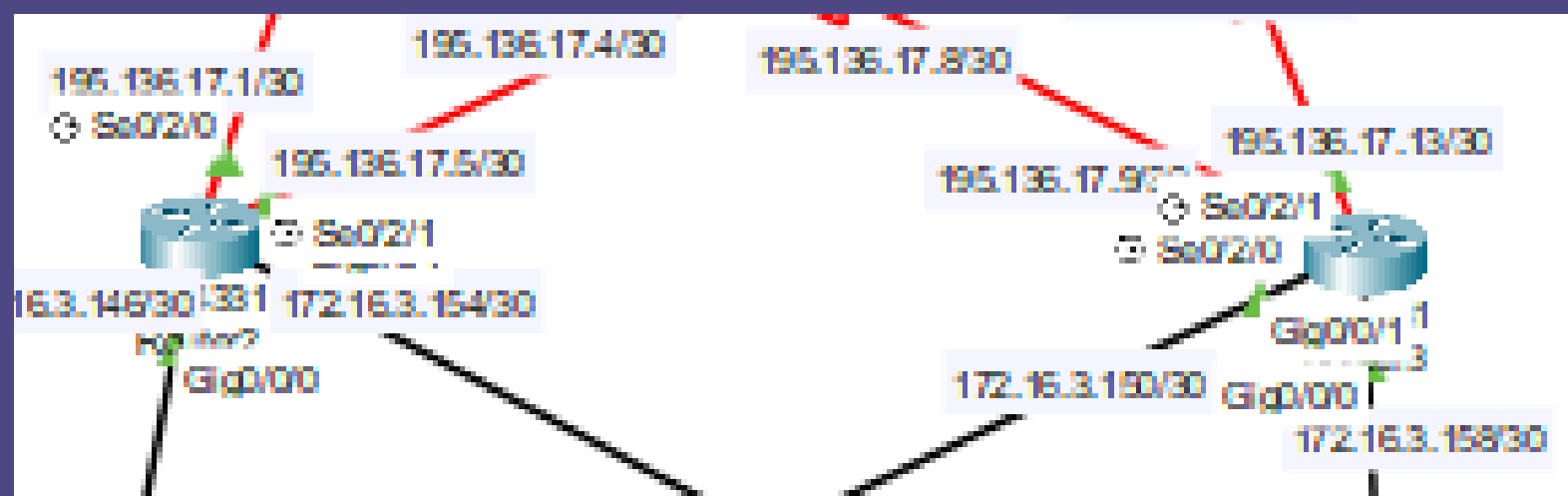
NETWORK DESCRIPTION

Physical design & Network Components:-

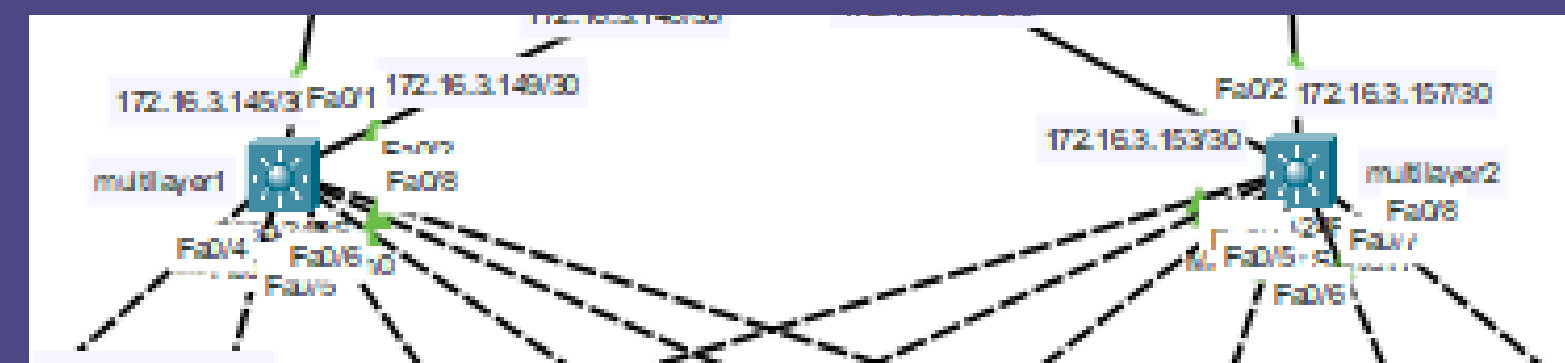
- The network is for a Company that is expected to have **three floors** , **two departments** in each floor.
- **Multilayer Switches:** Combining routing and switching.
- **Access Points:** For wireless networks.
- **Servers:** Hosting company services.
- **End Devices:** PCs, laptops, and mobile devices



Sales Department



Core Routers

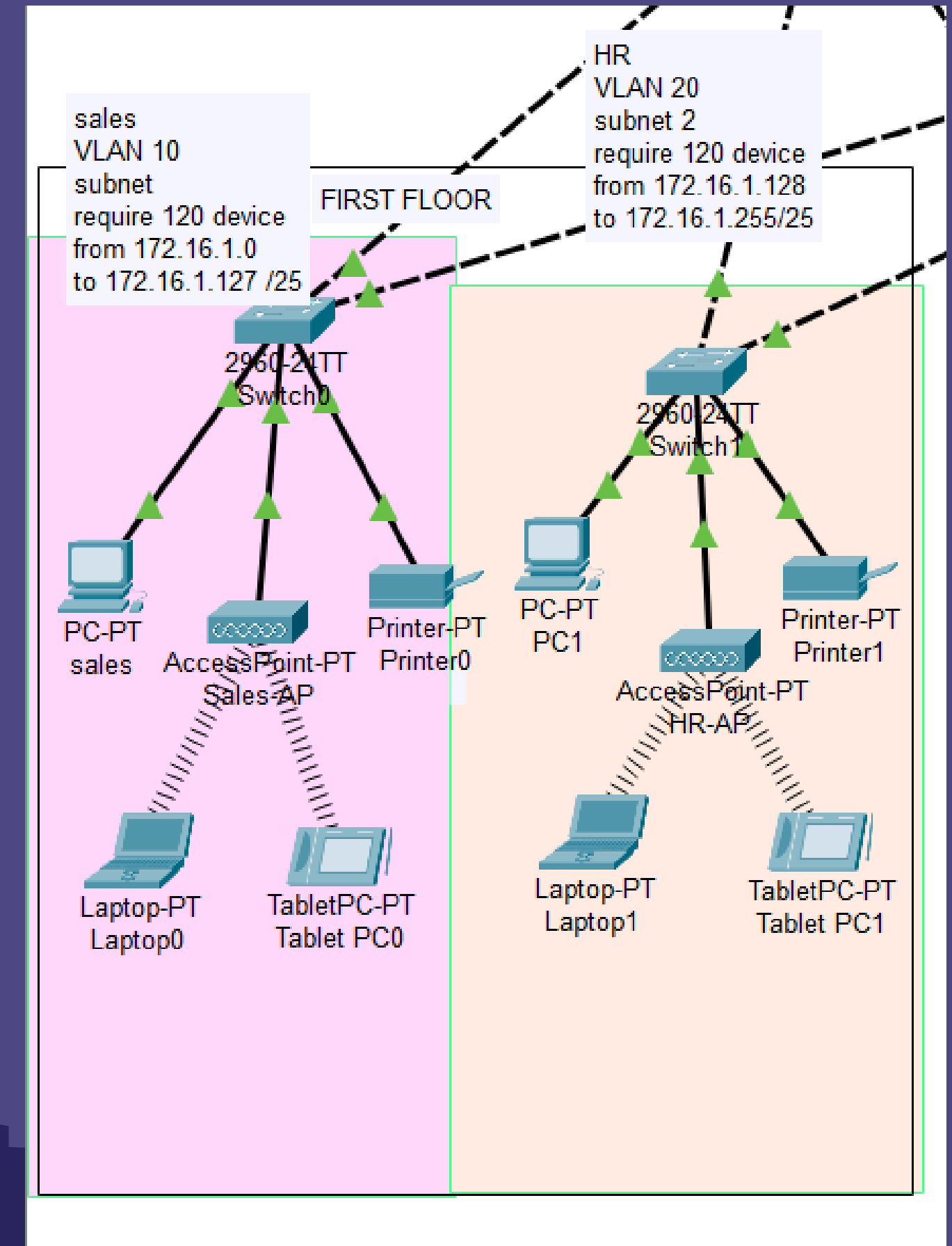


layer 3 switches

PHYSICAL DESIGN

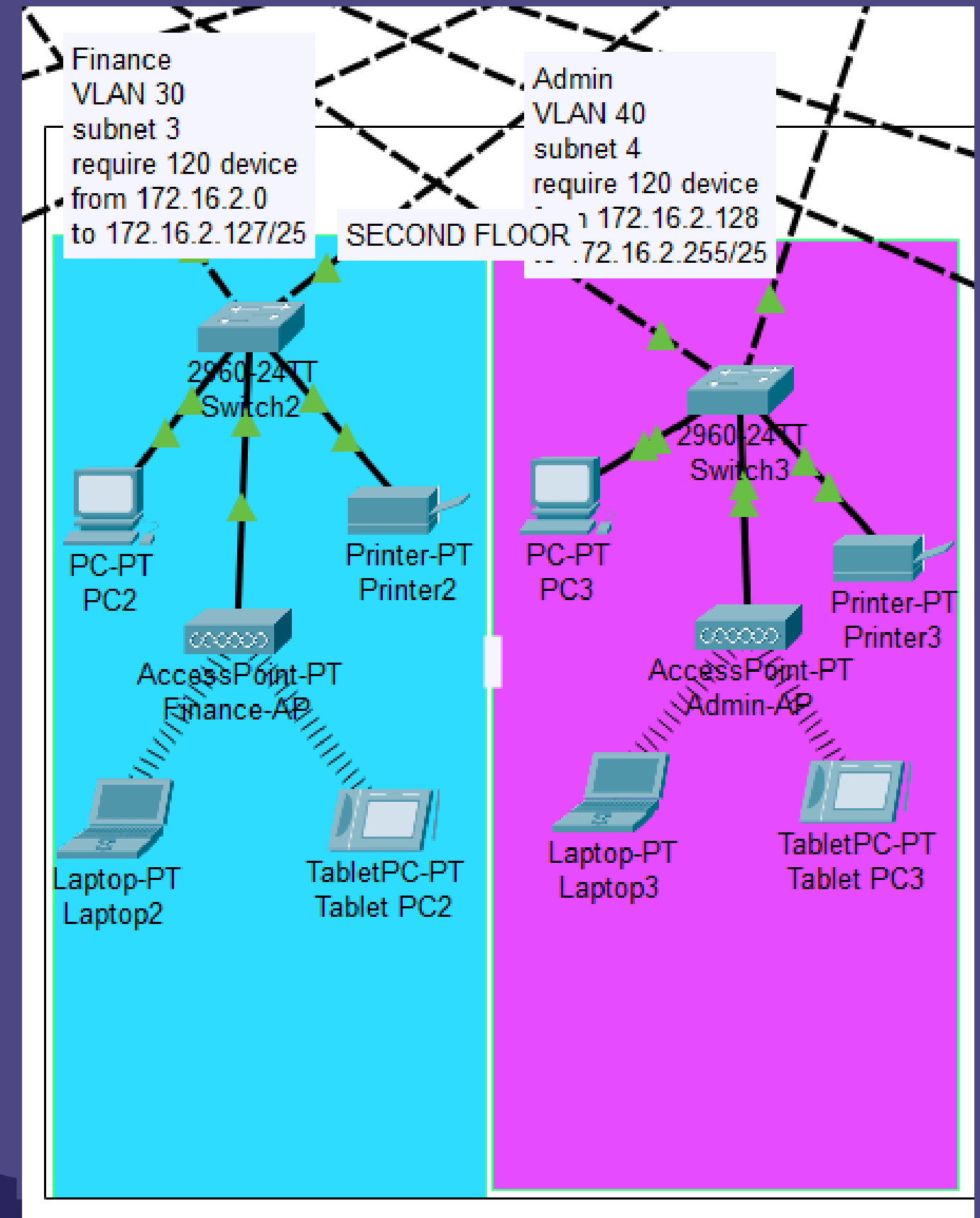
First floor

has **Sales department**, which is expected to have 120 employees, and **HR department** which is expected to have 120 users.



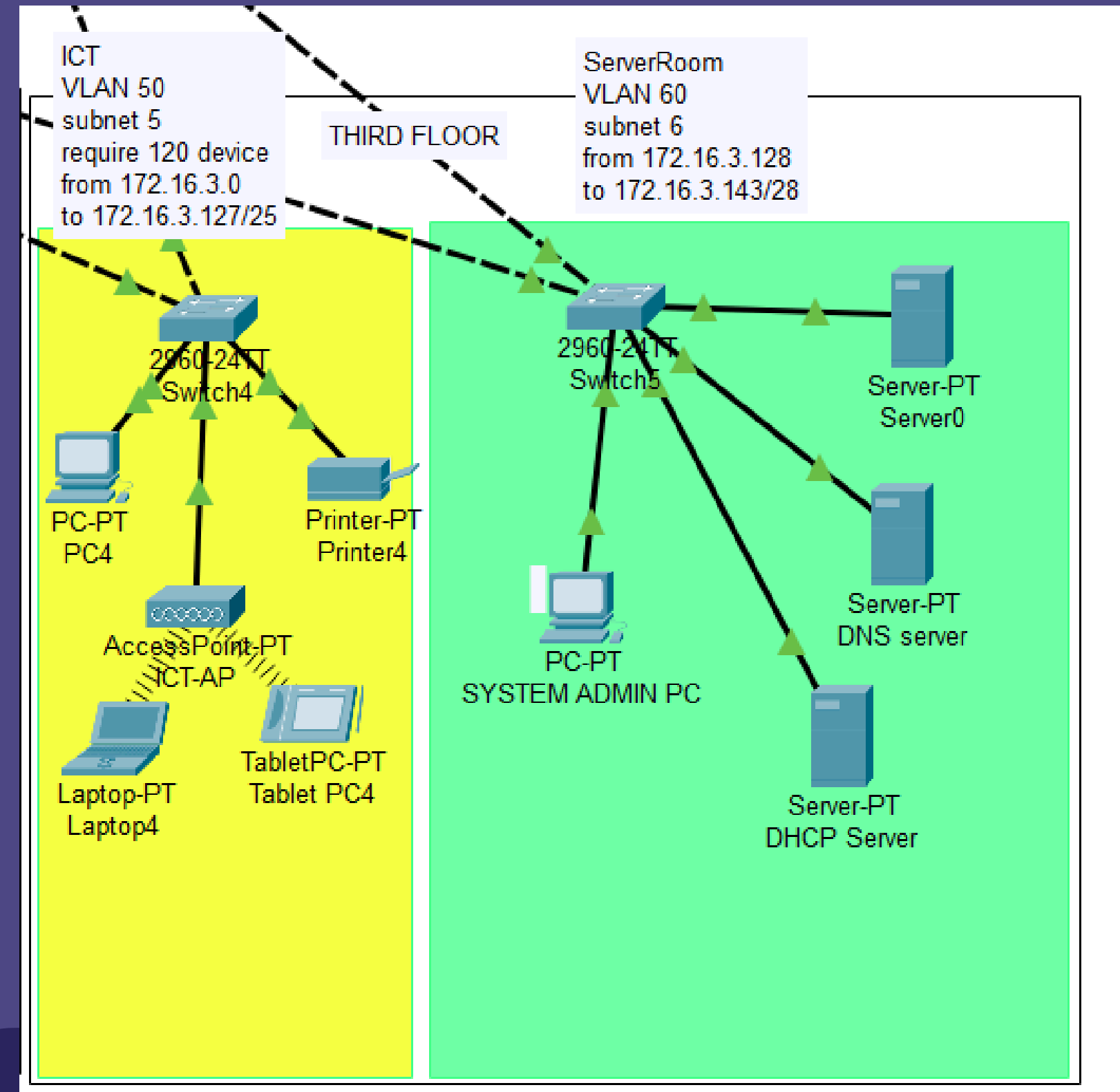
PHYSICAL DESIGN

Second floor has **Finance department** which is expected to have 120 employees and **Administrators and public relations department**, which is expected to have 120 employees.



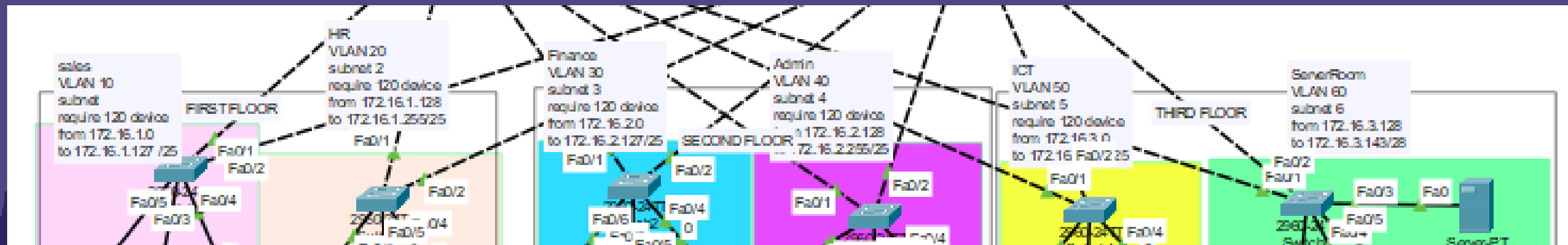
PHYSICAL DESIGN

Third floor has **ICT** which is expected to have 120 Employees, and **server room** can have 12 devices.



NETWORK CHARACTERISTICS

- Each department has a wireless network for the users.
- The base network of address **172.16.1.0** is subnetted to subnets for each department.
- The company network is connected to the IPs addressed:
195.136.17.0/30, 195.136.17.4/30, 195.136.17.8/30, 195.136.17.12/30
connected to the two ISPs (to make redundancy) .
- All the devices in the network obtained their IP addresses **dynamically** from **the DHCP servers** located in the server room. Devices in the server room obtained their IPs statically.



ADDING DEVICES & CONNECTIONS

- Two company routers connected to ISPs via Serial DTE.
- Access layer switches added for each department.
- Two multilayer switches link the routers to access switches.
- Admin PC and servers connected to the server room switch.
- Copper straight-through cables connect routers to distribution switches.
- Copper cross-over cables connect distribution to access switches.



BASIC CONFIGURATION

- Applying basic settings on all switches and routers.
- Basic settings include setting the hostname, login passwords, enable passwords services, and banner.

```
line con 0
  password core1
  login
!
```

```
hostname core1
!
!
!
enable secret 5 $1$mERr$8fa3z6U28TRsEX0invri00
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username core1 password 0 core1
```


IP ASSIGNMENT TO ROUTERS

- we assign ips to routers and layer 3 switches using CLI Command window

```
mutlilayer1(config-if)#int f0/1  
mutlilayer1(config-if)#no shut  
mutlilayer1(config-if)#ip add 172.16.3.145  
255.255.255.252
```

```
core1#show ip int br  
Interface                               IP-Address  
GigabitEthernet0/0/0                    172.16.3.146  
GigabitEthernet0/0/1                    172.16.3.154  
GigabitEthernet0/0/2                    unassigned  
GigabitEthernet0/1/0                    unassigned  
GigabitEthernet0/1/1                    unassigned  
GigabitEthernet0/1/2                    unassigned  
GigabitEthernet0/1/3                    unassigned  
Serial0/2/0                             195.136.17.1  
Serial0/2/1                             195.136.17.5
```



SUBNETTING AND ADDRESSING

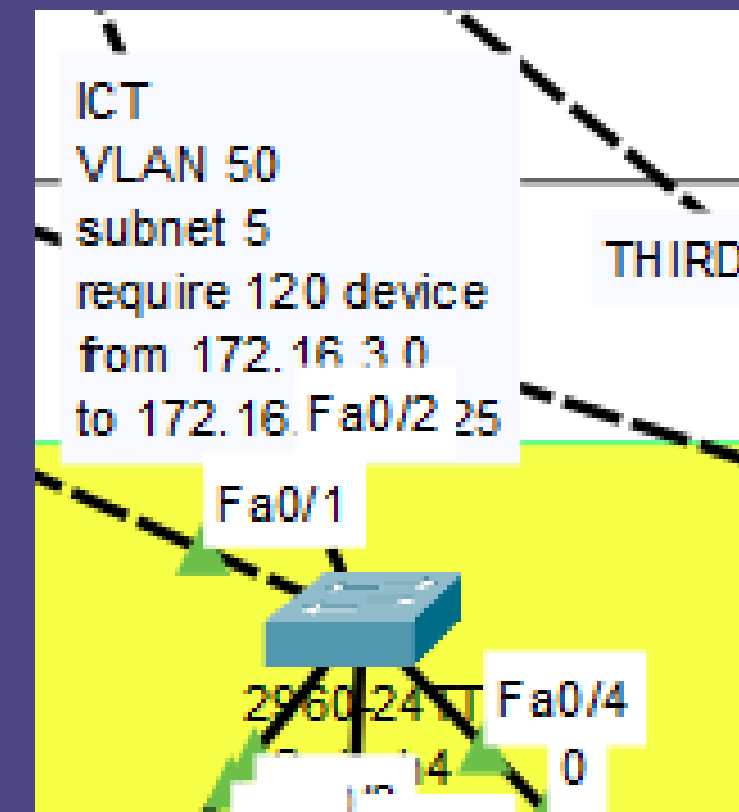
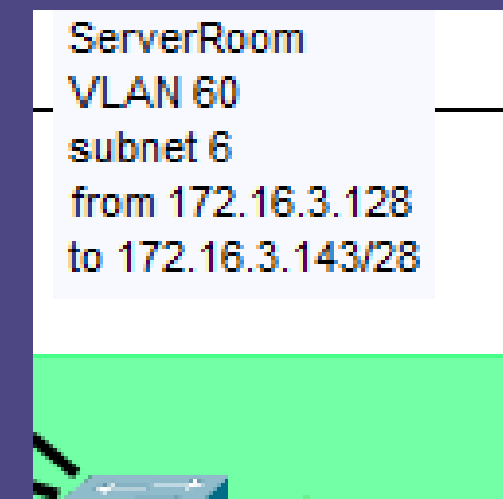
- For Ex The network 172.16.3.0 is divided into subnets.

IPv4 Address	172.16.3.6
Subnet Mask	255.255.255.128
Default Gateway	172.168.3.1

- PC takes ip address from the subnet ips range

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	172.16.1.12
Subnet Mask	255.255.255.128

Printer takes ip address from the same range of the subnet Ips



SSH CONFIGURATION

- SSH is setup in multilayer switches and the two main routers.
- The SSH version is set to 2.

```
Mlt-SW1(config)#crypto key generate rsa
The name for the keys will be: Mlt-SW1.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for you
  General Purpose Keys. Choosing a key modulus greater than 512 may ta
    a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Mlt-SW1(config)#line vty 0 15
^Mar 1 0:8:59.705: %SSH-5-ENABLED: SSH 1.99 has been enabled
Mlt-SW1(config-line)#login loc
Mlt-SW1(config-line)#login local
Mlt-SW1(config-line)#tran
Mlt-SW1(config-line)#transport in
Mlt-SW1(config-line)#transport input s
Mlt-SW1(config-line)#transport input ssh
Mlt-SW1(config-line)#ex
% Ambiguous command: "ex"
Mlt-SW1(config-line)#
Mlt-SW1(config-line)#exit
Mlt-SW1(config)#
Mlt-SW1(config)#
Mlt-SW1(config)#
Mlt-SW1(config)#do wr
Building configuration...
```


SSH IN NETWORK DESIGN

- **Secure Shell (SSH)** secures remote access to network devices using encryption.
- Protects data from unauthorized access and replaces insecure protocols like Telnet.

```
line vty 5 15
 login local
 transport input ssh
!
```

```
[Connection to 172.16.3.146 closed by foreign host]
C:\>ssh -l core1 172.16.3.146

Password:

hello from core1 sw ^_^

core1>en
Password:
core1#
core1#
core1#
core1#
```

VLANs CONFIGURATION

- VLANs are setup for every department switch mapped to the department switches.
- Trunk ports were configured for inter-switch communication.

```
HR-SW(config-if-range)#switchport mode trunk

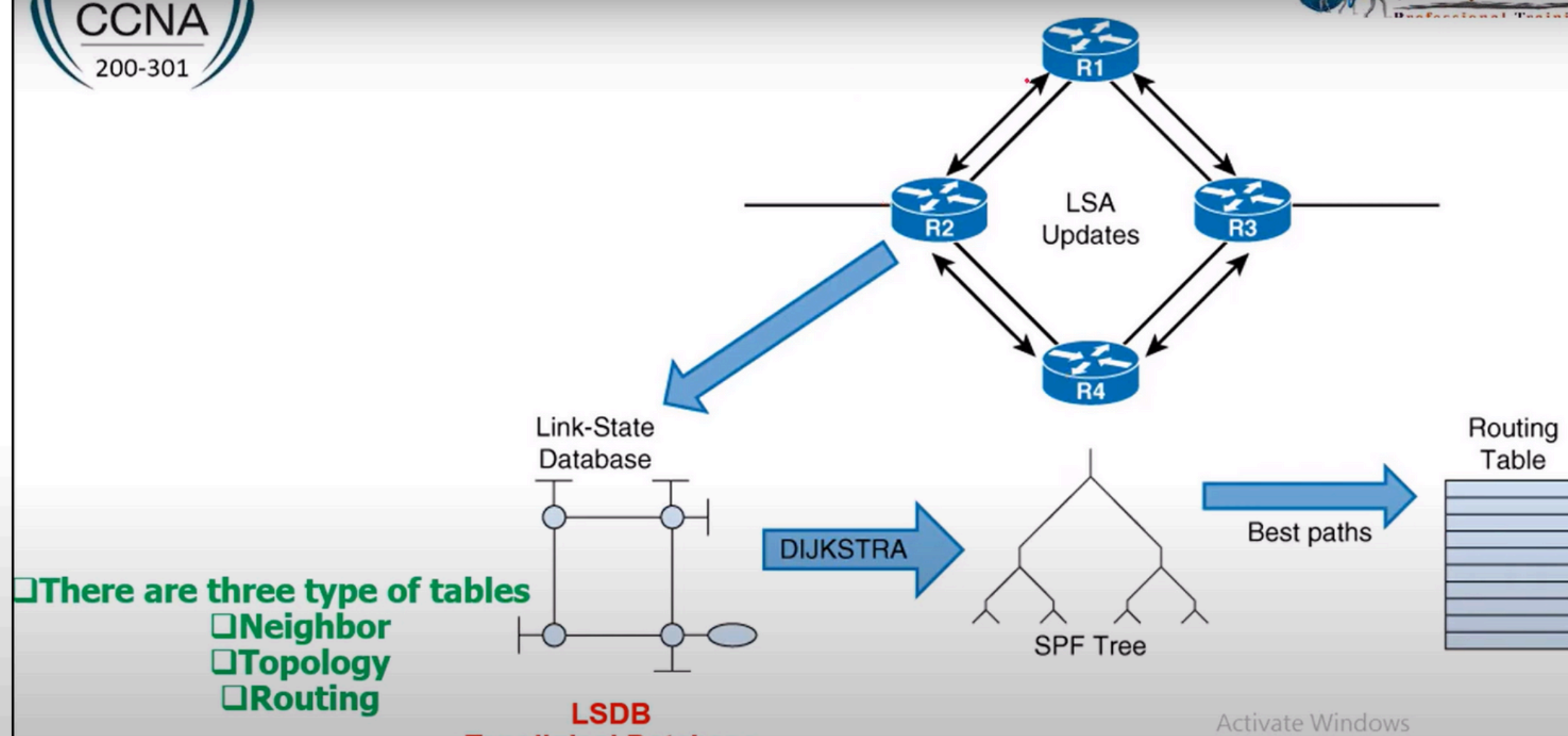
HR-SW(config-if-range)#exit
HR-SW(config)#
HR-SW(config)#vlan 20
HR-SW(config-vlan)#name HR
HR-SW(config-vlan)#vlan 99
HR-SW(config-vlan)#name BlackHole
HR-SW(config-vlan)#exit
HR-SW(config)#
HR-SW(config)#int range fa0/3-24
HR-SW(config-if-range)#switchport mode access
HR-SW(config-if-range)#switchport access vlan 20
HR-SW(config-if-range)#exit
HR-SW(config)#
HR-SW(config)#int range gig0/1-2
HR-SW(config-if-range)#switchport mode access
HR-SW(config-if-range)#switchport access vlan 99
HR-SW(config-if-range)#exit
HR-SW(config)#
HR-SW(config)#do wr
Building configuration...
[OK]
HR-SW(config)#
HR-SW(config)#
HR-SW(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

- ❑ Standard Protocol
- ❑ Classless Protocol
- ❑ Sent Packet 224.0.0.5
- ❑ Administrative Distance = 110
- ❑ Metric = Cost = $2^8 / \text{Bandwidth}$
- ❑ Loop Free Topology
- ❑ Unlimited Number Of Hop Count



OSPF

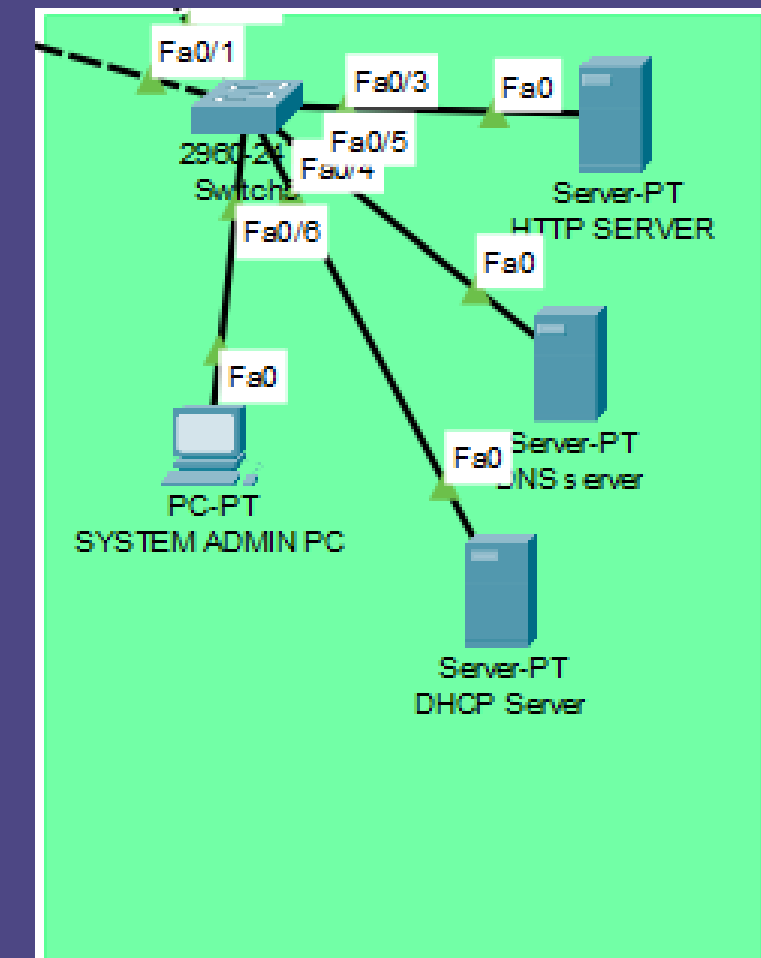
OSPF Adjacency

```
03:27:59: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on
FastEthernet0/1 from LOADING to FULL, Loading Done
```

```
03:27:59: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on
FastEthernet0/2 from LOADING to FULL, Loading Done
```

DHCP SERVER SETUP

- DHCP pool was first created.
- Then defined a network range and a default gateway.
- DHCP bindings on client devices were verified.



DHCP configuration interface showing the 'serverPool' configuration.

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Start IP Address: 172.16.3.128

Subnet Mask: 255.255.255.240

Maximum Number of Users: 512

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add, Save, Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
SalesPool	172.168.1.1	172.16.3.131	172.16.1.6	255.255.255.240	120	0.0.0.0	0.0.0.0
FinancePool	172.168.2.1	172.16.3.131	172.16.2.6	255.255.255.240	120	0.0.0.0	0.0.0.0
ICTPool	172.168.3.1	172.16.3.131	172.16.3.6	255.255.255.240	120	0.0.0.0	0.0.0.0
AdminPool	172.168.2....	172.16.3.131	172.16.2.134	255.255.255.240	120	0.0.0.0	0.0.0.0
HRPool	172.168.1....	172.16.3.131	172.16.1.134	255.255.255.240	120	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.3.128	255.255.255.240	512	0.0.0.0	0.0.0.0

DHCP SERVER

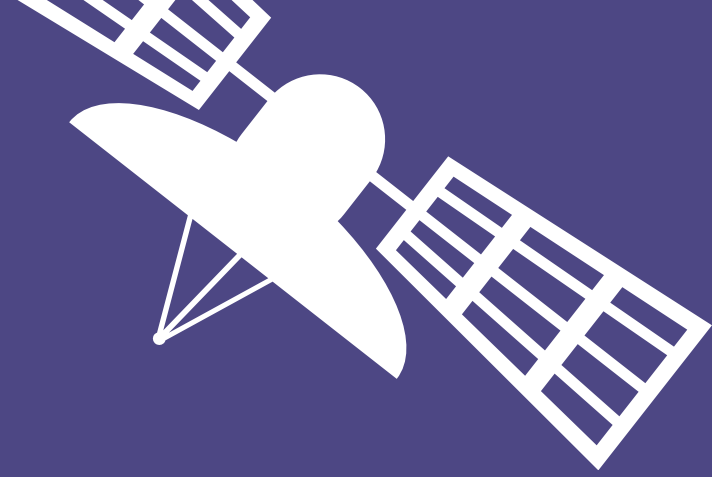
- here the pcs and the printers take a dynamic ip address from the DHCP server and DNS from the DNS Servers successfully

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	172.16.2.6
Subnet Mask	255.255.255.128

IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	172.16.2.8
Subnet Mask	255.255.255.128
Default Gateway	172.168.2.1
DNS Server	172.16.3.131



ROUTING TABLES



- Some Routing tables for routers and layer3 switches in our network

Routing table of core1

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
    172.16.0.0/16 is variably subnetted, 12 subnets, 4
masks
O    172.16.1.0/25 [110/2] via 172.16.3.145,
00:34:12, GigabitEthernet0/0/0
    [110/2] via 172.16.3.153,
00:34:12, GigabitEthernet0/0/1
O    172.16.1.128/25 [110/2] via 172.16.3.145,
00:34:12, GigabitEthernet0/0/0
    [110/2] via 172.16.3.153,
00:34:12, GigabitEthernet0/0/1
O    172.16.2.0/25 [110/2] via 172.16.3.145,
00:34:12, GigabitEthernet0/0/0
    [110/2] via 172.16.3.153,
00:34:12, GigabitEthernet0/0/1
O    172.16.2.128/25 [110/2] via 172.16.3.145,
00:34:12, GigabitEthernet0/0/0
    [110/2] via 172.16.3.153,
00:34:12, GigabitEthernet0/0/1
O    172.16.3.0/25 [110/2] via 172.16.3.145,
00:34:12, GigabitEthernet0/0/0
    [110/2] via 172.16.3.153,
00:34:12, GigabitEthernet0/0/1
O    172.16.3.128/28 [110/2] via 172.16.3.145,
00:34:12, GigabitEthernet0/0/0
```

Routing table of layer 3 sw-2

```
masks
C    172.16.1.0/25 is directly connected, Vlan10
C    172.16.1.128/25 is directly connected, Vlan20
C    172.16.2.0/25 is directly connected, Vlan30
C    172.16.2.128/25 is directly connected, Vlan40
C    172.16.3.0/25 is directly connected, Vlan50
C    172.16.3.128/28 is directly connected, Vlan60
O    172.16.3.144/30 [110/2] via 172.16.3.154,
04:02:09, FastEthernet0/1
O    172.16.3.148/30 [110/2] via 172.16.3.158,
04:02:09, FastEthernet0/2
C    172.16.3.152/30 is directly connected,
FastEthernet0/1
C    172.16.3.156/30 is directly connected,
FastEthernet0/2
    195.136.17.0/30 is subnetted, 4 subnets
O    195.136.17.0 [110/65] via 172.16.3.154,
04:02:09, FastEthernet0/1
O    195.136.17.4 [110/65] via 172.16.3.154,
04:02:09, FastEthernet0/1
O    195.136.17.8 [110/65] via 172.16.3.158,
04:02:09, FastEthernet0/2
O    195.136.17.12 [110/65] via 172.16.3.158,
04:02:09, FastEthernet0/2
S*   0.0.0.0/0 is directly connected, FastEthernet0/1
```

WIRELESS NETWORK CONFIGURATION

- Components: access points (APs), SSIDs
- APs are set up with proper SSIDs.
- Static IPs were assigned to APs.
- Encryption protocols like WPA2 were configured.

Port Status ☒ On

SSID

2.4 GHz Channel

Coverage Range (meters)

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key

PSK Pass Phrase

Link Information **Connect** Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
Finance-AP	1	49%
Sales-AP	1	49%
HR-AP	1	49%
Admin-AP	1	49%

Site Information

Wireless Mode Infrastructure
Network Type Mixed B/G
Radio Band Auto
Security WPA2-PSK
MAC Address 00E0.8FA1.ADC4

Refresh **Connect**

2.4GHz

Adapter is Active

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. WPC300N

NAT (PAT)

- Inside and outside interfaces were defined.
- IP NAT pools were configured.
- NAT rules applied to ACLs.
 - ACL controls traffic flow and enhance security.

```
core1# SHOW IP NA TR
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	195.136.17.5:1	172.16.1.137:1	195.136.17.2:1	195.136.17.2:1
icmp	195.136.17.5:2	172.16.1.137:2	195.136.17.2:2	195.136.17.2:2
icmp	195.136.17.5:3	172.16.1.137:3	195.136.17.2:3	195.136.17.2:3
icmp	195.136.17.5:4	172.16.1.137:4	195.136.17.2:4	195.136.17.2:4

```
CORE-R1(config)#ip nat inside s
CORE-R1(config)#ip nat inside source a
CORE-R1(config)#ip nat inside source ?
    list      Specify access list describing local addresses
    static    Specify static local->global mapping
CORE-R1(config)#ip nat inside source lis
CORE-R1(config)#ip nat inside source list 1 int se0/2/0 ov
CORE-R1(config)#ip nat inside source list 1 int se0/2/0 overload
CORE-R1(config)#ip nat inside source list 1 int se0/2/1 overload
CORE-R1(config)#
```


HTTP

HTTP

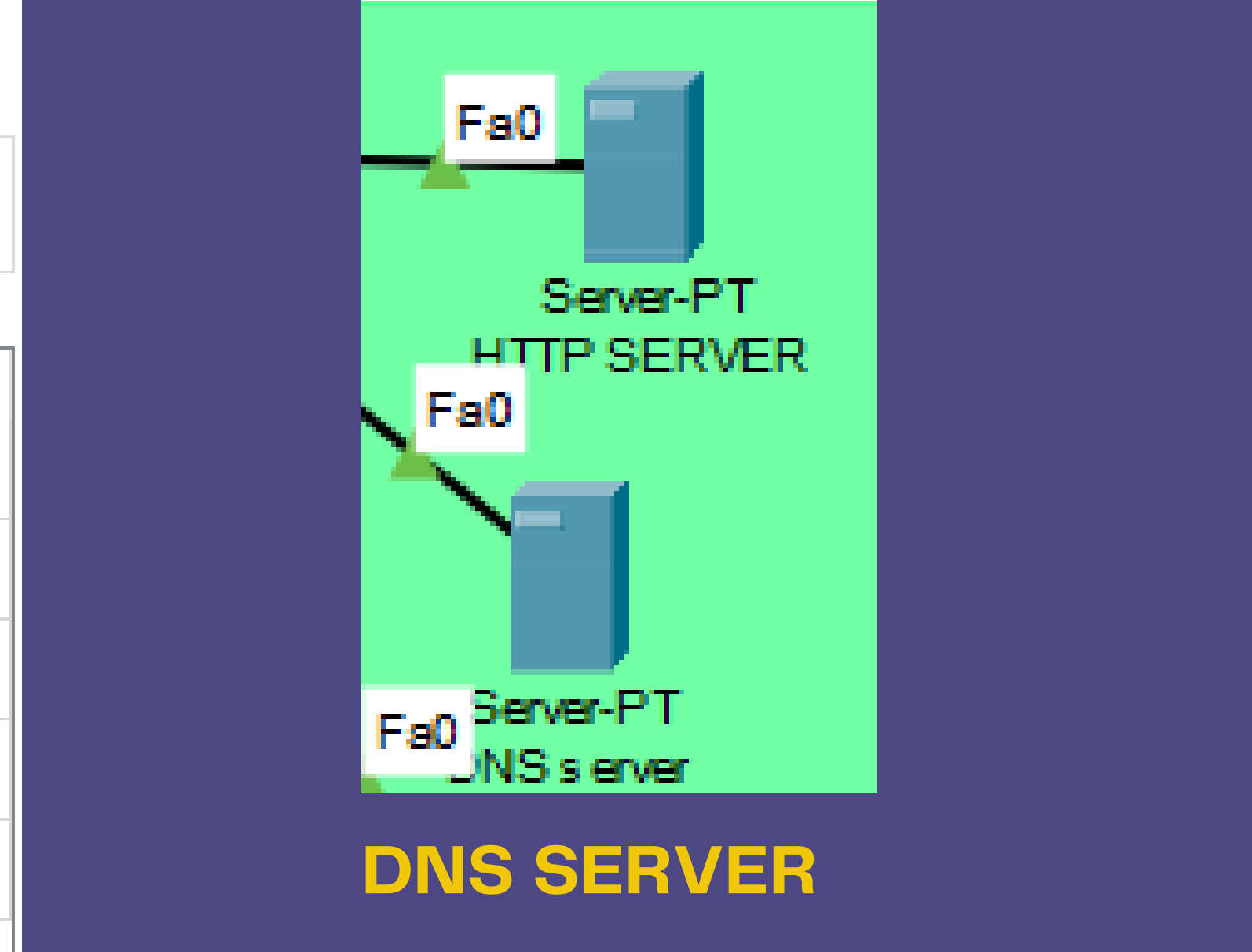
☒ On☐ Off

HTTPS

☒ On☐ Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)



DNS

DNS Service

☒ On☐ Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	www.gtech.com	A Record	172.16.3.131